A Working Review of h-Recursion in Two-Dimensional CFT

Virasoro, Neveu–Schwarz, and Ramond blocks

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Abstract

This note reviews the practical recipe for Zamolodchikov recursion in an internal conformal weight. The emphasis is on the data that a numerical implementation actually needs: the Kac pole, inverse null norm, factorized three-point matrix element, shifted module, elliptic or plumbing normalization, and regular large-weight part. We begin with Virasoro sphere and torus blocks, then summarize the $\mathcal{N} = 1$ NS and Ramond generalizations. The Ramond discussion makes explicit how the G_0 ground-state doublet is hidden in the scalar recursions of the literature. The final sections isolate the obstruction to applying the same method directly to a genus-two theta graph.

Contents

1 The question answered by h-recursion

Fix a chiral algebra, a pants decomposition, the central charge c, all external weights, and all internal weights except one, denoted h. A conformal-block coefficient obtained by descendant sewing is a rational function of h. Its singularities occur when the internal Verma module contains a null vector. The block can therefore be reconstructed from

$$\boxed{\text{regular part at large } h \quad + \quad \text{sum over Kac poles and their residues.}} \tag{1}$$

This is the meaning of h-recursion. The residue is local to the two vertices adjacent to the singular internal edge. The regular part is not local in this sense: it depends on the topology, channel, and normalization. This distinction will be central at genus two.

At fixed order in the plumbing parameters, only finitely many Kac labels contribute. Once the regular part is known, the recursion is triangular and efficient; it never requires construction or inversion of a large Gram matrix.

1.1 Three versions that should not be conflated

The same meromorphic idea appears in three commonly used forms:

1. A recursion for coefficients of a local coordinate expansion, such as the sphere z-series.
2. An elliptic recursion after extracting a universal prefactor and replacing z by a nome $q(z)$. This is usually the fastest numerical representation of a four-point block.
3. A plumbing recursion on a multipoint or higher-genus graph, with one plumbing parameter q_e for every internal edge.

The poles and null-vector residues are the same representation-theory data. The prefactor and regular part differ.

2 Universal residue mechanism

Let $\mathcal{V}_{c,h}$ be a Verma module with a basis $\{|h;I\rangle\}$ and Gram matrix

$$B^{(n)}_{IJ}(c,h) = \langle h;I|h;J\rangle, \qquad |I| = |J| = n. \tag{2}$$

A sewn block coefficient has the schematic form

$$F_n(c,h) = \sum_{|I|=|J|=n} \rho_L(I) \left(B^{(n)}(c,h)^{-1}\right)^{IJ} \rho_R(J). \tag{3}$$

Suppose that at $h = h_{r,s}(c)$ a normalized singular vector

$$|\chi_{r,s}\rangle = D_{r,s}|h_{r,s}\rangle \tag{4}$$

appears at level $\ell_{r,s}$. Descendants of this vector form a Verma submodule with highest weight

$$h'_{r,s} = h_{r,s} + \ell_{r,s}. \tag{5}$$

Near the zero of the Kac determinant, the inverse Gram matrix has a rank-one singular part. Define the inverse null-norm slope by

$$A_{r,s}(c) = \left[\lim_{h\to h_{r,s}} \frac{\langle\chi_{r,s}(h)|\chi_{r,s}(h)\rangle}{h - h_{r,s}}\right]^{-1}. \tag{6}$$

The Ward identities factorize the two adjacent three-point tensors:

$$\rho_L(D_{r,s}u) = P^{(L)}_{r,s}\,\rho'_L(u), \qquad\qquad \rho_R(D_{r,s}v) = P^{(R)}_{r,s}\,\rho'_R(v). \tag{7}$$

The $P_{r,s}$ are fusion polynomials, or finite matrices when the ground space is not one-dimensional. Equations (6)–(7) imply

$$\operatorname*{Res}_{h=h_{r,s}} \mathcal{F} = q^{\ell_{r,s}}\,A_{r,s}\,P^{(L)}_{r,s}P^{(R)}_{r,s}\,\mathcal{F}(h_{r,s}+\ell_{r,s}). \tag{8}$$

This is the reusable kernel of every h-recursion discussed below.

2.1 What must be fixed before quoting a residue

A residue coefficient is meaningless until the following choices are locked:

1. the normalization of the singular operator $D_{r,s}$;
2. whether $A_{r,s}$ is a slope with respect to h, a momentum, or another parameter;
3. the ordering and normalization of the two three-point forms;
4. the local coordinate or elliptic prefactor;
5. the branch convention for momenta such as b, P, or β.

A large fraction of apparent sign and factor-of-two disagreements in the literature come from changing one item in this list.

3 What is the regular boundary data?

3.1 Coefficient-level definition

At a fixed plumbing level N, suppose the reduced block coefficient is a meromorphic function of one recursion variable x:

$$H_N(x) = U_N(x) + \sum_p \frac{R_{p,N}}{x - x_p}. \tag{9}$$

The pole sum is fixed by null-vector factorization. The function U_N is the regular boundary data. If $H_N(x)$ is bounded as $x \to \infty$, then Liouville's theorem gives

$$U_N = \lim_{x\to\infty} H_N(x). \tag{10}$$

If H_N grows as a polynomial of degree k, then $U_N(x)$ is a polynomial of degree k, and $k+1$ asymptotic coefficients are needed. In practice one chooses a prefactor so that the reduced block is bounded and the regular part is as simple as possible.

At the level of the complete plumbing series,

$$\mathcal{H}_\infty(\boldsymbol{q}) = \sum_N \boldsymbol{q}^N U_N \tag{11}$$

is called the large-weight seed. Thus "the seed is one" means

$$U_0 = 1, \qquad U_{N>0} = 0 \tag{12}$$

for the reduced block, not for the original unstripped conformal block.

3.2 Why a prefactor is part of the boundary condition

The original block may contain a known exponential or character-like large-weight dependence:

$$\mathcal{F}_x(\boldsymbol{q}) = \mathcal{P}_x(\boldsymbol{q})\,\mathcal{H}_x(\boldsymbol{q}). \tag{13}$$

One first derives $\mathcal{P}_x$, then applies the partial-fraction argument to $\mathcal{H}_x$. Changing $\mathcal{P}_x$ changes the regular part. Consequently the statement

$$\lim_{x\to\infty} \mathcal{H}_x = 1 \tag{14}$$

is meaningful only together with the chosen elliptic or plumbing normalization.

3.3 How the large-weight limit is derived

There are two standard derivations.

Semiclassical monodromy and uniformization. For the sphere four-point block, Zamolodchikov studied a classical limit in which c, the internal weight, and the external weights are large with their ratios controlled. The monodromy problem determines the leading large-internal-weight dependence in terms of the elliptic nome. The answer is precisely the prefactor in Eq. (60); after it is removed, the reduced elliptic block tends to one. The pillow representation gives a geometric interpretation of the same factor.

Large-weight power counting in the descendant basis. At fixed c,

$$[L_n, L_{-n}] = 2nL_0 + \frac{c}{12}(n^3 - n) \sim 2nh \tag{15}$$

on a state of large weight h. For a PBW state $L_{-A}|h\rangle$, let $[A]$ be the number of Virasoro generators. After Gram–Schmidt orthogonalization,

$$\|\ell_{-A}|h\rangle\|^2 \sim h^{[A]}. \tag{16}$$

Off-diagonal inverse-Gram elements are suppressed. If a three-point tensor contains two large-weight legs and one fixed-weight primary, Ward identities allow the descendant generators to pass through the fixed primary at subleading cost. Hence

$$\frac{\rho(\ell_{-A}h, d, \ell_{-B}h)}{\|\ell_{-A}|h\rangle\| \, \|\ell_{-B}|h\rangle\|} \longrightarrow \delta_{AB}. \tag{17}$$

The boundary seed is then obtained by summing the descendant states that survive this diagonal limit.

3.4 The standard examples

Sphere four-point block. The original z-block has a nontrivial large-h behavior. After the elliptic prefactor is extracted, one has

$$\mathcal{H}_h(q) \longrightarrow 1. \tag{18}$$

The coefficientwise regular data are therefore $U_0 = 1$, $U_{N>0} = 0$.

Torus one-point block. In the trace, the large-h normalized matrix element of a fixed primary approaches the descendant norm. Every descendant therefore contributes one, and the unstripped boundary is the nondegenerate Verma character:

$$\mathcal{F}^d_h(q) \longrightarrow q^{h-c/24} \prod_{n=1}^{\infty} \frac{1}{1-q^n}. \tag{19}$$

After this character is divided out, the reduced seed is again one.

Torus N-point necklace channel. The regular part in one weight h_i is complicated. Rather than compute it directly, Cho–Collier–Yin set

$$h_i = H + a_i, \qquad a_1 = 0, \tag{20}$$

and take $H \to \infty$ with every difference $a_i = h_i - h_1$ fixed. Each vertex then has two large legs and one fixed external primary, so Eq. (17) applies around the entire necklace. All descendant partitions must agree around the cycle, giving

$$\mathcal{F}_{\text{necklace}} \longrightarrow \sum_{A} (q_1 q_2 \cdots q_N)^{|A|} = \prod_{n=1}^{\infty} \frac{1}{1-(q_1 q_2 \cdots q_N)^n}. \tag{21}$$

The block is regarded as a meromorphic function of the single common variable H. A Kac pole on edge i occurs at

$$H = h_{r,s} - a_i. \tag{22}$$

Thus all edge poles and one simultaneous boundary value determine a closed one-variable recursion. This avoids the need to know the separate regular functions $U_i(\{h_{j\neq i}\})$.

Sphere N-point linear channel. A similar one-parameter limit sends all internal weights and the two endpoint external weights to infinity with their differences fixed. Only the level-zero term survives, so

$$\mathcal{F}_{\text{linear}} \longrightarrow 1. \tag{23}$$

The price is that the recursion shifts some external weights as well as the internal weights [3].

3.5 Neveu–Schwarz necklace boundary at fixed weight differences

We now repeat the Cho–Collier–Yin simultaneous-weight argument for the $\mathcal{N}=1$ NS algebra, using the notation and BPZ conventions of the human note. This subsection treats only NS internal edges and bottom components of fixed NS external superprimaries. The Ramond ground doublet requires a separate argument and is not used below.

Block and spin-sign conventions. Write an NS descendant as

$$\mathbf{L}_{-\mathsf{A}}\phi_h = L_{-n_1}\cdots L_{-n_\ell} G_{-r_1}\cdots G_{-r_p}\phi_h, \qquad r_j \in \mathbb{Z}_{>0} + \frac{1}{2}, \tag{24}$$

where repeated bosonic modes are allowed and the fermionic modes are strictly ordered. Set

$$|\mathsf{A}| = \sum_j n_j + \sum_k r_k, \qquad p(\mathsf{A}) = p \pmod 2, \qquad \#\mathsf{A} = \ell + p. \tag{25}$$

For even internal ground states and normalized even three-point forms $\rho_0(\phi_{h_i}, \phi_{d_i}, \phi_{h_{i+1}}) = 1$, the descendant part of the NS necklace block is

$$\begin{aligned} \mathcal{F}^{\text{NS}}_{\boldsymbol{\eta}}(\boldsymbol{q}; \boldsymbol{h}, \boldsymbol{d}) = &\sum_{\{\mathsf{A}_i, \mathsf{B}_i\}} \prod_{i=1}^{N} \left[q_i^{|\mathsf{A}_i|} \eta_i^{p(\mathsf{A}_i)} (\mathbf{G}_{h_i})^{\mathsf{A}_i \mathsf{B}_i} \right] \\ &\times \prod_{i=1}^{N} \rho_0\left(\mathbf{L}_{-\mathsf{B}_i}\phi_{h_i}, \phi_{d_i}, \mathbf{L}_{-\mathsf{A}_{i+1}}\phi_{h_{i+1}}\right), \end{aligned} \tag{26}$$

with cyclic indices and $|\mathsf{A}_i| = |\mathsf{B}_i|$. Because every displayed vertex uses ρ_0, its two descendant levels are restricted to have integral sum, or equivalently $p(\mathsf{B}_i) = p(\mathsf{A}_{i+1})$. The factors $q_i^{h_i - c/24}$, which carry the primary propagation, are not included in Eq. (26). The plumbing signs $\eta_i = \pm 1$ act on a descendant precisely as in the human note. Define

$$H = h_1, \qquad h_i = H + a_i, \qquad a_1 = 0, \qquad Q = \prod_{i=1}^{N} q_i, \qquad \varepsilon = \prod_{i=1}^{N} \eta_i. \tag{27}$$

The large-weight limit below keeps c, d_i, a_i, q_i, η_i fixed and sends $H \to \infty$. It is understood coefficientwise in the plumbing series.

Large-H NS Gram matrix and orthogonal basis. The two diagonal brackets responsible for the leading norm are

$$[L_n, L_{-n}] = 2nL_0 + \mathcal{O}(H^0), \qquad \{G_r, G_{-r}\} = 2L_0 + \mathcal{O}(H^0). \tag{28}$$

Fix a level and a fermion-parity block of the NS Verma module. If the mode L_{-n} occurs m_n times, define

$$C_{\mathsf{A}} = 2^{\#\mathsf{A}} \prod_{n \geq 1} n^{m_n} m_n!. \tag{29}$$

The leading fermionic contraction contributes $2H$, with no additional power of its half-integral mode number. Repeated use of Eq. (28) gives the super-Gram power counting

$$(\mathbf{G}_{H+a})_{\mathsf{AB}} = \begin{cases} C_{\mathsf{A}} H^{\#\mathsf{A}}(1 + \mathcal{O}(H^{-1})), & \mathsf{A} = \mathsf{B}, \\ \mathcal{O}(H^{\#\mathsf{A}-1}), & \mathsf{A} \neq \mathsf{B}, \quad \#\mathsf{A} = \#\mathsf{B}, \\ \mathcal{O}\left(H^{\min(\#\mathsf{A}, \#\mathsf{B})}\right), & \#\mathsf{A} \neq \#\mathsf{B}. \end{cases} \tag{30}$$

This statement is within a fixed level and parity block. In particular, the second line says that two distinct PBW strings with the same oscillator number cannot contract every mode pair diagonally.

Order the PBW strings first by $\#\mathsf{A}$, and apply Gram–Schmidt within each fixed level and parity block. The resulting orthogonal state can be chosen in the form

$$\boldsymbol{\ell}_{-\mathsf{A}} \phi_h = \mathbf{L}_{-\mathsf{A}} \phi_h + \sum_{\substack{|\mathsf{B}| = |\mathsf{A}|,\, p(\mathsf{B}) = p(\mathsf{A}) \\ \#\mathsf{B} \leq \#\mathsf{A},\, \mathsf{B} \neq \mathsf{A}}} f^{\mathsf{A}}_{\mathsf{B}}(c, h)\, \mathbf{L}_{-\mathsf{B}} \phi_h. \tag{31}$$

This is the NS analogue of CCY (3.10). Equation (30) fixes the coefficient scaling,

$$f^{\mathsf{A}}_{\mathsf{B}}(c, H + a) = \begin{cases} \mathcal{O}(H^{-1}), & \#\mathsf{B} = \#\mathsf{A}, \quad \mathsf{B} \neq \mathsf{A}, \\ \mathcal{O}(H^0), & \#\mathsf{B} < \#\mathsf{A}, \end{cases} \tag{32}$$

which is the analogue of CCY (3.11). Its diagonal norm is

$$\mathcal{N}_{\mathsf{A}}(H + a) := \|\boldsymbol{\ell}_{-\mathsf{A}} \phi_{H+a}\|^2 = C_{\mathsf{A}} H^{\#\mathsf{A}} \left(1 + \mathcal{O}(H^{-1})\right). \tag{33}$$

This is the NS version of CCY (3.12).

The sewn block in the orthogonal basis. Since a_i is fixed, replacing $H + a_i$ by H in the leading Gram matrices and vertices changes only subleading terms. The inverse Gram matrices are diagonal in the orthogonal basis, so

$$\begin{aligned}\mathcal{F}^{\mathrm{NS}}_{\boldsymbol{\eta}} \longrightarrow & \sum_{\mathsf{A}_1,\ldots,\mathsf{A}_N} \prod_{i=1}^{N} q_i^{|\mathsf{A}_i|} \eta_i^{p(\mathsf{A}_i)} \\ & \times \prod_{i=1}^{N} \frac{\rho_0\big(\boldsymbol{\ell}_{-\mathsf{A}_i}\phi_H, \phi_{d_i}, \boldsymbol{\ell}_{-\mathsf{A}_{i+1}}\phi_H\big)}{\mathcal{N}_{\mathsf{A}_i}(H)}, \qquad \mathsf{A}_{N+1} = \mathsf{A}_1.\end{aligned} \tag{34}$$

This is the direct NS analogue of CCY (3.13), including the plumbing signs.

Power counting of the NS three-point tensor. Extend the definition in Eq. (31) by setting $f^{\mathsf{A}}_{\mathsf{A}} = 1$ and $f^{\mathsf{A}}_{\mathsf{C}} = 0$ for $\#\mathsf{C} > \#\mathsf{A}$. Then

$$\begin{aligned}&\rho_0(\boldsymbol{\ell}_{-\mathsf{A}}\phi_H, \phi_d, \boldsymbol{\ell}_{-\mathsf{B}}\phi_H) \\ &\quad = \sum_{\substack{|\mathsf{C}|=|\mathsf{A}|,\, p(\mathsf{C})=p(\mathsf{A}) \\ |\mathsf{D}|=|\mathsf{B}|,\, p(\mathsf{D})=p(\mathsf{B})}} f^{\mathsf{A}}_{\mathsf{C}} f^{\mathsf{B}}_{\mathsf{D}} \rho_0(\mathbf{L}_{-\mathsf{C}}\phi_H, \phi_d, \mathbf{L}_{-\mathsf{D}}\phi_H)\,.\end{aligned} \tag{35}$$

This is the NS counterpart of CCY (3.14).

Move the BPZ-conjugate positive modes on the first leg through the fixed external primary. A commutator with ϕ_d contains no factor of H; only a direct positive-negative mode contraction produces a factor of H. Consequently

$$\begin{aligned}&\rho_0(\mathbf{L}_{-\mathsf{C}}\phi_H, \phi_d, \mathbf{L}_{-\mathsf{D}}\phi_H) \\ &\quad = \rho_0\Big(\phi_H, \phi_d, (\mathbf{L}_{-\mathsf{C}})^{\dagger}\mathbf{L}_{-\mathsf{D}}\phi_H\Big) + \text{terms with at least one external-field commutator} \\ &\quad = \mathcal{O}\Big(H^{\min(\#\mathsf{C},\#\mathsf{D})}\Big)\,.\end{aligned} \tag{36}$$

This is precisely the superconformal analogue of CCY (3.15): the anticommutator $\{G_r, G_{-r}\}$ supplies the same one power of H as a bosonic contraction. If the two oscillator numbers are equal but the PBW labels differ, the leading term is an off-diagonal Gram element and is one power lower, as in the second line of Eq. (30).

Combining Eqs. (32), (33), (35), and (36) gives

$$\rho_0(\boldsymbol{\ell}_{-\mathsf{A}}\phi_H, \phi_d, \boldsymbol{\ell}_{-\mathsf{B}}\phi_H) = \begin{cases} \mathcal{N}_{\mathsf{A}}(H)\big(1 + \mathcal{O}(H^{-1})\big), & \mathsf{A} = \mathsf{B}, \\ \mathcal{O}(H^{\#\mathsf{A}-1}), & \mathsf{A} \neq \mathsf{B}, \quad \#\mathsf{A} = \#\mathsf{B}, \\ \mathcal{O}\Big(H^{\min(\#\mathsf{A},\#\mathsf{B})}\Big)\,, & \#\mathsf{A} \neq \#\mathsf{B}. \end{cases} \tag{37}$$

In particular,

$$\frac{\rho_0(\boldsymbol{\ell}_{-\mathsf{A}}\phi_H, \phi_d, \boldsymbol{\ell}_{-\mathsf{A}}\phi_H)}{\mathcal{N}_{\mathsf{A}}(H)} \longrightarrow 1. \tag{38}$$

Equation (38) is the exact NS analogue of CCY (3.16). For $\mathsf{A} \neq \mathsf{B}$, division by $\sqrt{\mathcal{N}_{\mathsf{A}}\mathcal{N}_{\mathsf{B}}}$ tends to zero. Therefore every surviving term in Eq. (34) obeys $\mathsf{A}_1 = \cdots = \mathsf{A}_N$.

Sum over the surviving NS states. A common bosonic oscillator of mode n contributes Q^n, while an occupied common fermionic oscillator of mode $n-\frac{1}{2}$ contributes $\varepsilon Q^{n-1/2}$. Hence

$$\boxed{\lim_{\substack{H\to\infty\\ a_i \text{ fixed}}} \mathcal{F}^{\mathrm{NS}}_{\boldsymbol{\eta}} = \chi^{(\varepsilon)}_{\mathrm{NS}}(Q) := \prod_{n=1}^{\infty} \frac{1+\varepsilon Q^{n-1/2}}{1-Q^n}.} \tag{39}$$

Thus $\varepsilon = +1$ gives the ordinary NS descendant character and $\varepsilon = -1$ gives the $(-1)^F$-twined character. Dividing the block by $\chi^{(\varepsilon)}_{\mathrm{NS}}(Q)$ produces a reduced block with unit large-H seed.

No pole at $H = \infty$. The power-counting limit must also exclude a polynomial ambiguity in the partial-fraction expansion. Fix a plumbing multi-level $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_N)$ and define

$$F_{\boldsymbol{\nu}}(H; \boldsymbol{a}) := [q_1^{\nu_1} \cdots q_N^{\nu_N}]\, \mathcal{F}^{\mathrm{NS}}_{\boldsymbol{\eta}}(H; \boldsymbol{a}). \tag{40}$$

At fixed $\boldsymbol{\nu}$, every descendant space is finite-dimensional. The Gram entries and three-point matrix elements are polynomial in the weights, while the inverse Gram entries are rational. Therefore $F_{\boldsymbol{\nu}}(H; \boldsymbol{a})$ is a rational function of H. For generic fixed a_i, its finite poles are the Kac poles

$$H = h^{\mathrm{NS}}_{r,s} - a_i, \qquad \frac{rs}{2} \le \nu_i, \tag{41}$$

with $r + s$ even.

Equations (34)–(39) prove coefficientwise that

$$F_{\boldsymbol{\nu}}(H; \boldsymbol{a}) = \chi^{(\varepsilon)}_{\boldsymbol{\nu}} + o(1), \qquad |H| \longrightarrow \infty, \tag{42}$$

where $\chi^{(\varepsilon)}_{\boldsymbol{\nu}}$ is the corresponding coefficient of $\chi^{(\varepsilon)}_{\mathrm{NS}}(Q)$. Explicitly it vanishes unless $\nu_1 = \cdots = \nu_N$. In the local coordinate $z = H^{-1}$, Eq. (42) says that the rational function $F_{\boldsymbol{\nu}}(z^{-1}; \boldsymbol{a})$ is bounded at $z = 0$. Hence $z = 0$, equivalently $H = \infty$, is a removable singularity and not a pole.

To see directly what this implies for the recursion, subtract all finite principal parts at this fixed plumbing level:

$$R_{\boldsymbol{\nu}}(H) = F_{\boldsymbol{\nu}}(H; \boldsymbol{a}) - \sum_{p} \frac{\operatorname{Res}_{H=H_p} F_{\boldsymbol{\nu}}(H; \boldsymbol{a})}{H - H_p}. \tag{43}$$

The function $R_{\boldsymbol{\nu}}$ has no finite poles. Every subtracted principal part is $\mathcal{O}(H^{-1})$ at infinity, while Eq. (42) shows that $R_{\boldsymbol{\nu}}$ has no pole there either. Thus $R_{\boldsymbol{\nu}}$ is holomorphic on the Riemann sphere and must be the constant

$$R_{\boldsymbol{\nu}}(H) = \chi^{(\varepsilon)}_{\boldsymbol{\nu}}. \tag{44}$$

This proves, rather than assumes, that there is no positive-power polynomial in H hidden in the regular part. The statement needed for the h-recursion is coefficientwise; a limit of the fully summed plumbing series would additionally require a uniform-convergence estimate.

What is the regular part? It is essential to make the partial-fraction decomposition only after the change of variables in Eq. (27). For generic fixed a_i, the block is a meromorphic function of the single variable H, with poles

$$H = h^{\mathrm{NS}}_{r,s} - a_i, \qquad r + s \in 2\mathbb{Z}, \qquad \ell_{r,s} = \frac{rs}{2}. \tag{45}$$

Coefficientwise in $\boldsymbol{q}$, its decomposition is

$$\mathcal{F}_{\boldsymbol{\eta}}^{\mathrm{NS}}(H;\boldsymbol{a}) = \chi_{\mathrm{NS}}^{(\varepsilon)}(Q) + \sum_{i=1}^{N} \sum_{\substack{r,s\geq 1\\ r+s\in 2\mathbb{Z}}} \frac{B_{i;r,s}^{\mathrm{NS}}(\boldsymbol{a})}{H + a_i - h_{r,s}^{\mathrm{NS}}}. \tag{46}$$

The residues $B_{i;r,s}^{\mathrm{NS}}$ are independent of H because all neighboring weights and the shifted block are evaluated at $H = h_{r,s}^{\mathrm{NS}} - a_i$. For $i \neq 1$, the null-submodule shift becomes

$$H \longrightarrow h_{r,s}^{\mathrm{NS}} - a_i, \qquad a_i \longrightarrow a_i + \ell_{r,s}, \tag{47}$$

whereas a pole on the base edge gives

$$H \longrightarrow h_{r,s}^{\mathrm{NS}} + \ell_{r,s}, \qquad a_j \longrightarrow a_j - \ell_{r,s} \quad (j = 2, \ldots, N). \tag{48}$$

When rs is odd, the null vector is odd and the residue changes the fermion-parity routing of the component block. The recursion should therefore be regarded as a finite vector recursion in parity-routing space: the trivial closed routing has the character in Eq. (39) as its regular entry, while non-diagonal routings have zero boundary value in the normalization above.

Equation (46) is not the single-edge decomposition at fixed $h_{j\neq i}$. If an old residue coefficient is first restricted to $h_j = H + a_j$, its apparent H-dependence splits into its value at the pole plus a term regular in H; the latter belongs to the new regular part. The character is the regular part of the fixed-$\boldsymbol{a}$ problem, not the separate function $U_i(\{h_{j\neq i}\})$.

Two-edge low-level check. For $N = 2$, the human-note Ward identities and $\|G_{-1/2}\phi_h\|^2 = 2h$ give

$$[q_1^{1/2} q_2^{1/2}]\mathcal{F}_{\boldsymbol{\eta}}^{\mathrm{NS}} = \eta_1\eta_2 \frac{(h_1 + h_2 - d_1)(h_1 + h_2 - d_2)}{4h_1 h_2} \longrightarrow \varepsilon. \tag{49}$$

An excitation on only one edge instead gives

$$[q_1]\mathcal{F}_{\boldsymbol{\eta}}^{\mathrm{NS}} = \frac{(h_1 + d_1 - h_2)(h_1 + d_2 - h_2)}{2h_1} \longrightarrow 0, \tag{50}$$

$$[q_2]\mathcal{F}_{\boldsymbol{\eta}}^{\mathrm{NS}} = \frac{(h_2 + d_1 - h_1)(h_2 + d_2 - h_1)}{2h_2} \longrightarrow 0. \tag{51}$$

These are the first nontrivial consequences of the diagonal selection rule: the character begins $1 + \varepsilon Q^{1/2} + Q + \cdots$, with no isolated $q_i^{1/2}$ or q_i term.

The argument assumes generic a_i, so that distinct Kac hyperplanes do not collapse to the same H-pole, and is a statement about every fixed plumbing coefficient rather than a uniform large-H limit of the summed series.

Ramond sector deferred. For R internal edges the positive modes obey analogous power counting, but the two-dimensional ground space and unsuppressed G_0 action remain in the leading vertex. One must keep those ground indices open before deciding which scalar components reduce to an R character. That matrix-valued problem will be treated separately rather than inferred from the NS result.

3.6 Why the theta graph is different

The successful limits above arrange that every trivalent tensor contains *two* large legs and *one fixed* primary. This makes the leading tensor a two-point norm and leads to Eq. (17).

On a genus-two theta graph, a simultaneous large-weight limit makes all three legs of each trinion large. The leading three-point tensor is then a genuine trivalent oscillator vertex rather than a diagonal two-point norm. The Kac residues do not determine this vertex. To obtain a closed h-recursion one must derive and resum the resulting oscillator network. For an NRR vertex the network also retains the Ramond ground fibre and the G_0 sewing matrices.

This gives a concrete criterion for a new h-recursion:

$$\boxed{\begin{array}{c}\text{find a one-parameter correlated large-weight limit,}\\ \text{prove that its trivalent large-weight tensor has a closed form,}\\ \text{and factor it so that the reduced block has a bounded seed.}\end{array}} \tag{52}$$

If no such closed trivalent seed is available, the local h-pole formulas remain valid but do not form a complete algorithm.

4 Virasoro four-point blocks on the sphere

Use

$$c = 1 + 6Q^2, \qquad Q = b + b^{-1}, \qquad h(\lambda) = \frac{Q^2 - \lambda^2}{4}. \tag{53}$$

The degenerate weights are

$$h_{r,s} = \frac{Q^2 - (rb + sb^{-1})^2}{4}, \qquad r, s \in \mathbb{Z}_{>0}, \tag{54}$$

and the singular vector is at level $\ell_{r,s} = rs$.

For the channel $\langle \phi_4(\infty)\phi_3(1)\phi_2(z)\phi_1(0)\rangle$, normalize

$$\mathcal{F}_h(z) = z^{h-h_1-h_2}\left(1 + \sum_{n\geq 1} z^n F_n(h)\right). \tag{55}$$

The first coefficient,

$$F_1(h) = \frac{(h + h_2 - h_1)(h + h_3 - h_4)}{2h}, \tag{56}$$

already displays the mechanism: a pole at the level-one degenerate weight $h = 0$, a factor from each adjacent three-point function, and a shifted module.

4.1 Coefficient recursion

At each level,

$$F_n(h) = U_n \sum_{\substack{r,s\geq 1\\ rs\leq n}} \frac{R_{r,s}}{h - h_{r,s}}\, F_{n-rs}(h_{r,s} + rs), \tag{57}$$

where

$$R_{r,s} = A_{r,s}\, P_{r,s}(h_4, h_3)\, P_{r,s}(h_2, h_1). \tag{58}$$

The h-independent term U_n is fixed by the large-h asymptotic. In a local z-series it is nontrivial; the elliptic normalization packages the complete sequence U_n into a universal prefactor.

4.2 Elliptic normalization

Introduce

$$q(z) = \exp\left[-\pi \frac{K(1-z)}{K(z)}\right]. \tag{59}$$

Zamolodchikov's normalization writes

$$\begin{aligned} \mathcal{F}_h(z) = (16q)^{h-(c-1)/24} z^{(c-1)/24-h_1-h_2} (1-z)^{(c-1)/24-h_2-h_3} \\ \times \theta_3(q)^{(c-1)/2-4\sum_i h_i} \mathcal{H}_h(q). \end{aligned} \tag{60}$$

The reduced block has the recursion

$$\boxed{\mathcal{H}_h(q) = 1 + \sum_{r,s\geq 1} \frac{(16q)^{rs} R_{r,s}}{h - h_{r,s}} \mathcal{H}_{h_{r,s}+rs}(q).} \tag{61}$$

The essential achievement of the elliptic prefactor is

$$\lim_{h\to\infty} \mathcal{H}_h(q) = 1. \tag{62}$$

The recursion is efficient because a requested coefficient of q^N receives contributions only from $rs \leq N$.

4.3 Practical implementation

To compute through q^N:

1. enumerate (r, s) with $rs \leq N$;
2. evaluate $h_{r,s}$, $A_{r,s}$, and the two fusion polynomials;
3. recursively evaluate the shifted block only through order $N - rs$;
4. memoize by shifted weight and remaining order;
5. multiply by the prefactor (60).

The recursion tree grows much more slowly than the descendant basis.

5 Virasoro torus one-point and multipoint blocks

For a torus one-point block,

$$\mathcal{F}_h^d(q) = q^{h-c/24} \sum_{n\geq 0} q^n \sum_{|I|=|J|=n} \rho(h; I, d, h; J)\, (B_h^{(n)})^{IJ}. \tag{63}$$

The same Kac pole appears on the traced module. Both sides of the singular state meet the same vertex, so the residue again contains two null-vector three-point factors:

$$\operatorname*{Res}_{h=h_{r,s}} \mathcal{F}_h^d = q^{rs} A_{r,s} P_{r,s}^{(1)}(d, h'_{r,s}) P_{r,s}^{(2)}(d, h_{r,s}) \mathcal{F}_{h'_{r,s}}^d. \tag{64}$$

After extracting the torus large-h character, the reduced block has seed one and satisfies the same triangular pattern as (61); see Ref. [2].

5.1 Linear and necklace channels

For a sphere multipoint block in a linear channel or a torus multipoint block in a necklace channel, every internal edge has two well-defined adjacent vertices. Applying (8) edge by edge gives

$$\operatorname*{Res}_{h_e=h_{r,s}} \mathcal{F}_\Gamma = q_e^{rs} A_{r,s} P_{r,s}^{(v_L,e)} P_{r,s}^{(v_R,e)} \mathcal{F}_\Gamma(h_e \to h_{r,s}+rs). \tag{65}$$

For these channels the simultaneous large-weight limit can be organized recursively, and closed h-recursions are available [3].

5.2 Why a local residue does not automatically give a closed recursion

For several internal weights, meromorphy in one variable gives

$$\mathcal{F}_\Gamma(\boldsymbol{h}) = U_e(\{h_{e'} : e' \neq e\}) + \sum_{r,s} \frac{\mathcal{R}_{r,s}^{(e)}}{h_e - h_{r,s}} \mathcal{F}_\Gamma(h_e \to h_{r,s}+rs). \tag{66}$$

The residue is known, but the regular function U_e need not be. At a theta vertex all three legs are internal. Sending one weight to infinity changes a three-point tensor whose other two weights remain dynamical. The large-h_e limit is therefore not the same universal external-leg asymptotic that closes the sphere linear and torus necklace recursions. This is the basic higher-genus obstruction.

Cho–Collier–Yin solve the arbitrary Virasoro plumbing problem instead by recursing in c. Its large-c regular part factorizes into a global $\mathfrak{sl}(2)$ block and a universal vacuum contribution. This is why the existence of graph-local h-pole residues should not be confused with the existence of a complete genus-two h-recursion.

6 $\mathcal{N}=1$ Neveu–Schwarz recursion

Use the ordinary super-Virasoro central charge

$$c = \frac{3}{2} + 3Q^2, \qquad Q = b + b^{-1}, \qquad h(\lambda) = \frac{Q^2-\lambda^2}{8}. \tag{67}$$

An NS singular vector is labelled by

$$h_{r,s}^{\mathrm{NS}} = \frac{Q^2-(rb+sb^{-1})^2}{8}, \qquad r+s \in 2\mathbb{Z}, \qquad \ell_{r,s} = \frac{rs}{2}. \tag{68}$$

Because levels may be half-integral, an NS block has even and odd superconformal components. A null vector of odd parity maps an even block to an odd block and conversely.

6.1 Component form

Let $\pi \in \{\mathrm{e},\mathrm{o}\}$ label the parity of the sewn three-point structure. The elliptic recursion takes the schematic form

$$\mathcal{H}_h^\pi(q) = g^\pi(q) + \sum_{\substack{r,s\geq 1 \\ r+s \text{ even}}} \frac{q^{rs/2} A_{r,s}^{\mathrm{NS}} P_{r,s}^{(L,\pi)} P_{r,s}^{(R,\pi)}}{h - h_{r,s}^{\mathrm{NS}}} \mathcal{H}^{\pi+\mathrm{parity}(rs/2)}_{h_{r,s}^{\mathrm{NS}}+rs/2}(q). \tag{69}$$

The precise powers of 16, signs, and seed functions g^π depend on the chosen four-point component and elliptic prefactor [4, 6].

The conceptual changes relative to Virasoro recursion are only:

1. the Kac level is $rs/2$;
2. NS fusion polynomials have even/odd versions;
3. the recursion can mix the two component blocks;
4. the large-h prefactor contains bosonic and fermionic oscillator characters.

6.2 Toric NS blocks

The same residue mechanism applies to torus one-point blocks. The trace can include either 1 or $(-1)^F$, producing the NS and $\widetilde{\mathrm{NS}}$ spin structures. The two blocks differ by signs for half-integral levels, while the Kac data are unchanged [7].

7 Ramond recursion: where the ground-state mixing goes

7.1 Long Ramond ground space

Parameterize a generic Ramond weight by

$$h_{\mathrm{R}}(\beta) = \frac{c}{24} - \beta^2. \tag{70}$$

In a parity basis (w_β^+, w_β^-),

$$G_0 w_\beta^\pm = ie^{\mp i\pi/4}\beta\, w_\beta^\mp, \qquad G_0^2 = -\beta^2 = h_{\mathrm{R}} - \frac{c}{24}. \tag{71}$$

After rephasing w_β^-, this is equivalently

$$G_0 = \beta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad (-1)^F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{72}$$

At $\beta = 0$ the long module shortens. One should perform the recursion at generic β and construct the irreducible short quotient separately.

7.2 Ramond Kac poles

The positive-parity singular state occurs at

$$\beta_{r,s} = \frac{rb + sb^{-1}}{2\sqrt{2}}, \qquad r + s \in 2\mathbb{Z} + 1, \qquad \ell_{r,s} = \frac{rs}{2}. \tag{73}$$

Its null submodule is another long Ramond module with

$$\beta'_{r,s} = \frac{(-1)^s}{2\sqrt{2}}\left(rb - sb^{-1}\right), \qquad h_{\mathrm{R}}(\beta'_{r,s}) = h_{\mathrm{R}}(\beta_{r,s}) + \frac{rs}{2}. \tag{74}$$

The odd null partner is generated by G_0:

$$\chi_{r,s}^- = \frac{e^{i\pi/4}}{i\beta'_{r,s}}\, G_0 \chi_{r,s}^+. \tag{75}$$

7.3 Why the published h-recursion is usually scalar

The toric construction of Ref. [7] starts with w_β^+ but allows G_0 in the descendant strings. It separates descendants into even and odd fermion parity and defines

$$F_{\beta,\mathrm{e}}^{\lambda(\pm)}, \qquad F_{\beta,\mathrm{o}}^{\lambda(\pm)}. \tag{76}$$

Here e/o is descendant parity, while $(\pm)$ labels the two independent chiral R–R–NS three-point forms. Ward identities give

$$F_{\beta,\mathrm{e}}^{\lambda(\pm)} = \pm F_{\beta,\mathrm{o}}^{\lambda(\pm)}, \tag{77}$$

so it is enough to recurse two scalar blocks. The 2×2 ground mixing has not disappeared: it has been diagonalized into these component relations.

7.4 The Ramond h-recursion

The natural meromorphic variable is β, with poles at $\beta = \pm\beta_{r,s}$. Combining the two poles gives a denominator

$$h_{\mathrm{R}}(\beta) - h_{r,s}^{\mathrm{R}} = -(\beta - \beta_{r,s})(\beta + \beta_{r,s}). \tag{78}$$

After extracting the appropriate large-β prefactor, the toric recursion is

$$\boxed{H_{\beta,\mathrm{e}}^{\lambda(\pm),f} = \delta_{f0} + \sum_{\substack{1<rs\leq 2f \\ r+s \text{ odd}}} \frac{A_{r,s}^{\mathrm{R}} P_c^{r,s}\begin{bmatrix}\Delta_\lambda \\ \pm\beta'_{r,s}\end{bmatrix} P_c^{r,s}\begin{bmatrix}\Delta_\lambda \\ \pm\beta_{r,s}\end{bmatrix}}{h_{\mathrm{R}}(\beta) - h_{r,s}^{\mathrm{R}}} H_{\beta'_{r,s},\mathrm{e}}^{\lambda(\pm),f-rs/2}.} \tag{79}$$

The phases from G_0, the odd null partner, and the two ground states are encoded in the two fusion polynomials and the $(\pm)$ label.

7.5 Blocks with an effective G_0 insertion

An odd Ramond plumbing modulus gives

$$e^{\alpha G_0} = \mathbf{1} + \alpha G_0, \qquad \int d\alpha\, e^{\alpha G_0} = G_0. \tag{80}$$

When the G_0 insertion can be moved to an adjacent vertex by a Ward identity, it becomes a starred or odd chiral component. Its leading large-β behavior contains an explicit factor of β. In the HJS toric conventions one finds schematically

$$\mathcal{F}_\beta^{*\,\lambda(-)} = e^{i\pi/4}\sqrt{2}\,\beta\, q^{h_{\mathrm{R}}-c/24}\chi_{\mathrm{R}}(q)\, \mathcal{H}_\beta^{\lambda(-)}(q). \tag{81}$$

After the β factor is extracted, the same reduced elliptic block obeys the scalar recursion (79). This is how known low-topology h-recursions deal with the G_0 matrix element.

7.6 External Ramond fields versus an internal Ramond edge

These two situations must be distinguished:

1. A four-R sphere block in the $R \times R$ channel propagates an NS module. Its recursion uses NS Kac poles together with R–R–NS fusion polynomials.

2. A block with an internal R module has the paired β poles, shifted momentum $\beta'_{r,s}$, and ground doublet described above.

The first problem tests Ramond external Ward identities but not the Ramond propagator.

8 A practical h-recursion checklist

For any proposed block, proceed in the following order.

Step 1: specify the block

- Choose the pants decomposition and local coordinates.
- Label every internal edge by Virasoro, NS, or R.
- Fix the ordered three-point form at every vertex.
- For R edges, fix parity basis, orientation, and spin lift.

Step 2: identify the meromorphic variable

- Virasoro or NS: normally the internal weight h.
- Ramond: preferably β, pairing the poles $\pm\beta_{r,s}$.

Step 3: derive the pole kernel

$$\text{kernel} = \text{inverse null slope} \times \text{left fusion factor} \times \text{right fusion factor}. \tag{82}$$

For an open R ground fibre, the last two factors are matrices and their order matters.

Step 4: determine the shifted block

Sector	Kac level	Shift
Virasoro	rs	$h \to h_{r,s} + rs$
NS	$rs/2$	$h \to h^{\mathrm{NS}}_{r,s} + rs/2$
R	$rs/2$	$\beta \to \beta'_{r,s}$

Step 5: derive the regular part

This is mandatory. A list of poles is not yet a recursion. One must derive, rather than guess,

$$\mathcal{F}_\infty = \lim_{\text{internal weight}\to\infty} (\text{properly reduced block})\,. \tag{83}$$

For standard sphere and torus elliptic blocks this limit is one. For multipoint and higher-genus blocks it may be a nontrivial function or finite-dimensional matrix.

Step 6: truncate by plumbing degree

At total plumbing degree N, retain only poles whose null level does not exceed the remaining degree. Memoization should use the full shifted weight data, component label, spin lift, and remaining multidegree.

Step 7: perform collision-safe evaluation

At rational or self-dual values of b, distinct labels can give the same pole. Terms must first be combined algebraically as Laurent or partial-fraction objects. Evaluating each resonant term separately and then adding generally loses the finite part.

9 Common failure modes

Failure	Diagnosis
Poles without a seed	The result is only a residue identity, not a closed recursion.
Using z where a formula assumes q	Powers of $16q$, theta functions, and the large-weight seed become incorrect.
Confusing $A^{(h)}_{r,s}$ and $A^{(\beta)}_{r,s}$	The two differ by the Jacobian $-1/(2\beta_{r,s})$.
Discarding the NS parity label	Odd null vectors map even and odd component blocks into one another.
Treating the R ground doublet as two unrelated modules	G_0 relates the states and fixes relative phases.
Projecting the R ground fibre too early	Spin-lift, G_0, and GSO matrices may not commute; orientation signs are lost.
Substituting $\beta = 0$ into a long-module formula	The Ramond module shortens and requires an irreducible quotient.
Calling a chiral block a correlator	The nonchiral pairing, structure constants, momentum integrals, spin sum, and string measure remain to be included.

10 Genus two: what the review tells us

10.1 The theta graph

A genus-two vacuum surface can be obtained by gluing two trinions along three edges. In an $\mathcal{N} = 1$ theory the chiral sector assignments are

$$NNN \qquad \text{or} \qquad NRR \tag{84}$$

at each vertex. For an NRR theta graph, denote the weights by

$$h_N, \qquad \beta_1, \qquad \beta_2 \tag{85}$$

and the plumbing parameters by $q_N, q_{R,1}, q_{R,2}$.

The edge-local h-pole data are known:

$$\begin{aligned} \operatorname*{Res}_{h_N = h^{\rm NS}_{r,s}} \mathcal{F}_\Theta &= q_N^{rs/2}\, \mathcal{R}^{\rm NS}_{r,s}\, \mathcal{F}_\Theta(h_N \to h^{\rm NS}_{r,s} + rs/2), \\ \operatorname*{Res}_{\beta_a = \pm\beta_{r,s}} \mathcal{F}_\Theta &= q_{R,a}^{rs/2}\, \mathbf{P}^{(a)}_{L,r,s}\, \mathcal{F}_\Theta(\beta_a \to \beta'_{r,s})\, \mathbf{P}^{(a)}_{R,r,s}. \end{aligned} \tag{86}$$

The Ramond formula is matrix-valued until the two odd sewings are contracted with

$$G_0^{(1)} \otimes G_0^{(2)}. \tag{87}$$

10.2 The missing ingredient

The unknown object is the simultaneous regular part,

$$\mathbf{U}_\Theta = \text{regular part of } \mathcal{F}_\Theta(h_N, \beta_1, \beta_2) \text{ in the internal weights.} \tag{88}$$

The scalar torus relations $F_{\mathrm{e}} = \pm F_{\mathrm{o}}$ cannot simply be applied independently to both R edges, because the two R lines meet at the same ordered R–R–NS tensor. One must first derive the joint Ward identities and the resulting finite matrix of component blocks.

Consequently,

$$\boxed{\text{known Ramond } h\text{-pole kernel } \not\Rightarrow \text{ known genus-two Ramond } h\text{-recursion.}} \tag{89}$$

10.3 Two plausible routes

Route A: derive a genuine multi-weight h-seed. Choose a basis of R–R–NS component tensors, extract the explicit $\beta_1\beta_2$ from the two G_0 sewings, and derive the joint large-(h_N, β_1, β_2) asymptotic. If the result resums to a known oscillator determinant, the local residues (86) close a matrix-valued h-recursion.

Route B: use genus-two c-recursion. Keep h_N, β_1, β_2 fixed while varying c, so that $G_0^2 = -\beta_a^2$ remains finite. The Kac residues are again edge-local. The remaining problem is a large-c NRR seed, which should follow from the contracted super-Virasoro oscillator algebra. This route is the closer analogue of the arbitrary-genus Virasoro construction of Ref. [3].

10.4 Fixed-β Ramond c-poles

The fixed-β formulation deserves to be separated from a fixed-h_R c-recursion. Write

$$h_R(c, \beta) = \frac{c}{24} - \beta^2 \tag{90}$$

and vary c with β held fixed. Then

$$G_0^2 = -\beta^2, \qquad q^{h_R - c/24} = q^{-\beta^2} \tag{91}$$

are independent of c. In particular, the level-zero denominator $(h_R - c/24)^{-1}$ is simply $-\beta^{-2}$; it does not produce an additional pole at $c = 24h_R$.

A positive-level Ramond pole occurs when

$$\beta = \epsilon\beta_{r,s}(b), \qquad \beta_{r,s}(b) = \frac{rb + sb^{-1}}{2\sqrt{2}}, \qquad \epsilon = \pm 1, \qquad r + s \text{ odd.} \tag{92}$$

Equivalently, the pole value b_* solves

$$rb_*^2 - \epsilon\, 2\sqrt{2}\,\beta b_* + s = 0, \qquad c^R_{r,s}(\beta) = \frac{3}{2} + 3\left(b_* + b_*^{-1}\right)^2. \tag{93}$$

The quadratic branches and the labels related by $b \leftrightarrow b^{-1}$, $r \leftrightarrow s$ must be organized once and continued consistently so that the same c-pole is not counted twice.

The denominator in the Ramond h-recursion becomes, along the fixed-β path,

$$h_R(c, \beta) - h^R_{r,s}(c) = \beta_{r,s}(c)^2 - \beta^2. \tag{94}$$

10.4.1 Residue recipe: first the null module, then the c-plane

It is useful to separate two logically distinct operations. The representation-theoretic residue is first determined in a local weight–momentum coordinate. It is then pulled back to the fixed-β c-plane.

At level f, a block coefficient has the schematic Gram-matrix form

$$F_f = \sum_{I,J} \rho_L(\nu_I)\,[B_f(c,\beta)^{-1}]^{IJ}\,\rho_R(\nu_J). \tag{95}$$

At $\beta = \epsilon\beta_{r,s}$, the primitive Ramond null multiplet $\chi^{\pm}_{r,s}$ appears at level $\ell_{r,s} = rs/2$. Only the inverse Gram matrix restricted to the descendant submodule generated by this multiplet is singular. Thus it is more precise to say that the primitive null direction creates the pole, while all descendants of the null multiplet resum into a block for the shifted Ramond module.

Choose the normalization of the singular-vector operator $D_{r,s}$ once and define its inverse-norm coefficient by

$$A^{(h),R}_{r,s} = \lim_{h\to h^R_{r,s}} (h - h^R_{r,s})\,\langle D_{r,s}w \mid D_{r,s}w\rangle^{-1}. \tag{96}$$

The three-point matrix elements of $D_{r,s}w$ factor into left and right Ramond fusion matrices,

$$\rho_L(D_{r,s}w) = \mathbf{P}_{L,r,s}\rho_L(w_{\beta'_{r,s}}), \qquad \rho_R(D_{r,s}w) = \rho_R(w_{\beta'_{r,s}})\mathbf{P}_{R,r,s}. \tag{97}$$

Keeping the two Ramond ground indices open automatically includes both $\chi^+_{r,s}$ and its G_0-related partner $\chi^-_{r,s}$.

Published Ramond kernel. The data above are not new. They were worked out for four-point blocks with R intermediate states in Ref. [6] and for toric one-point blocks in Ref. [7]. In the normalization

$$D_{r,s} = L_{-1}^{rs/2} + \cdots, \tag{98}$$

the weight-residue normalization is

$$A^{(h),R}_{r,s} = 2^{rs-2}\left(rb + \frac{s}{b}\right) \prod_{\substack{m=1-r,\dots,r\\ n=1-s,\dots,s\\ m+n\in 2\mathbb{Z}\\ (m,n)\neq(0,0)}} \left(mb + nb^{-1}\right)^{-1}. \tag{99}$$

The coefficient used for a pole in β in Ref. [6] already includes the first Jacobian:

$$A^{(\beta),R}_{r,s} = -\frac{A^{(h),R}_{r,s}}{2\beta_{r,s}} = -2^{rs-\frac{3}{2}} \prod_{\substack{m=1-r,\dots,r\\ n=1-s,\dots,s\\ m+n\in 2\mathbb{Z}\\ (m,n)\neq(0,0)}} \left(mb + nb^{-1}\right)^{-1}. \tag{100}$$

Because $r+s$ is odd, the factor $(m,n) = (r,s)$ is absent from the even sublattice automatically, and the two expressions obey exactly $A^{(\beta),R}_{r,s} = -A^{(h),R}_{r,s}/(2\beta_{r,s})$. The odd-sublattice condition printed below the corresponding product in Ref. [6] is a typo: it fails already for the exact level-one 2×2 Ramond Gram matrix. The even sublattice reproduces the residues at both $\beta_{2,1}$ and $\beta_{1,2}$.

For an adjacent NS weight

$$h_N = \frac{Q^2}{8} - \frac{\lambda^2}{8}, \qquad Q = b + b^{-1}, \tag{101}$$

define

$$X_{\mathrm{e}}^{r,s}(x) = \prod_{\substack{p=1-r+2k,\ q=1-s+2l \\ 0\le k\le r-1,\ 0\le l\le s-1 \\ k+l \text{ even}}} \frac{x - pb - qb^{-1}}{2\sqrt{2}},$$

$$X_{\mathrm{o}}^{r,s}(x) = \prod_{\substack{p=1-r+2k,\ q=1-s+2l \\ 0\le k\le r-1,\ 0\le l\le s-1 \\ k+l \text{ odd}}} \frac{x - pb - qb^{-1}}{2\sqrt{2}}, \tag{102}$$

$$P_{c,\sigma}^{r,s}(\beta;\lambda) = X_{\mathrm{e}}^{r,s}(2\sqrt{2}\beta - \sigma\lambda)\, X_{\mathrm{o}}^{r,s}(2\sqrt{2}\beta + \sigma\lambda), \qquad \sigma = \pm 1. \tag{103}$$

These are the two Ramond fusion-polynomial components denoted $P_c^{rs}[{}^{\Delta_\lambda}_{\pm\beta}]$ in the toric paper.

In the component basis of the four-point paper, the residue at $\beta = \epsilon\beta_{r,s}$ takes the universal form

$$\operatorname*{Res}_{\beta=\epsilon\beta_{r,s}} \mathbf{F} = \epsilon\, A_{r,s}^{(\beta),R}\, \mathbf{P}_{L,r,s}^{(\epsilon)}\, \mathbf{F}(\beta'_{r,s})\, \mathbf{P}_{R,r,s}^{(\epsilon)}, \qquad \epsilon = \pm 1. \tag{104}$$

Here each $\mathbf{P}^{(\epsilon)}$ is selected from (103) with the component and orientation signs specified by the corresponding R–R–NS three-point form. Equation (104), including these sign changes, is the local kernel to import for every Ramond plumbing edge.

At fixed c, the same denominator may be resolved into the two momentum poles:

$$\frac{1}{h_R(\beta) - h_{r,s}^R} = -\frac{1}{(\beta - \beta_{r,s})(\beta + \beta_{r,s})}, \qquad \operatorname*{Res}_{\beta=\epsilon\beta_{r,s}} = -\frac{1}{2\epsilon\beta_{r,s}}. \tag{105}$$

In the c-recursion, however, β itself is not varied. The local coordinate at the pole is

$$u_{r,s}(c) = \epsilon\beta_{r,s}(c) - \beta, \qquad u_{r,s}(c_*) = 0, \tag{106}$$

and

$$\beta_{r,s}(c)^2 - \beta^2 = 2\beta\, u_{r,s}(c) + O(u_{r,s}^2). \tag{107}$$

Consequently, the conversion from the momentum residue to the c-plane is entirely the Jacobian of this change of local coordinate.

Therefore the pole-conversion Jacobian is

$$\begin{aligned} J_{r,s}^R(\beta) &= \left[\frac{\partial\, \beta_{r,s}(c)^2}{\partial c}\right]_{c=c_{r,s}^R(\beta)}^{-1} \\ &= \left.\frac{dc/db}{d\beta_{r,s}^2/db}\right|_{b=b_*} = \left.\frac{24(b+b^{-1})(1-b^{-2})}{(rb+sb^{-1})(r-sb^{-2})}\right|_{b=b_*}. \end{aligned} \tag{108}$$

Equivalently, if one imports the published β-residue coefficient (100) directly, then

$$J_{r,s}^R A_{r,s}^{(h),R} = -\left.\frac{A_{r,s}^{(\beta),R}}{\partial_c \beta_{r,s}}\right|_{c=c_{r,s}^R(\beta)}. \tag{109}$$

This identity is the complete modification required to turn the known Ramond h-recursion pole kernel into a fixed-β c-plane kernel; it does not require a new singular-vector calculation.

With ground indices kept open, the corresponding edge residue is

$$\operatorname*{Res}_{c=c_{r,s}^R(\beta_e)} \mathbf{F}_\Gamma = q_e^{rs/2}\, J_{r,s}^R(\beta_e)\, A_{r,s}^{(h),R}\, \mathbf{P}_{L,r,s}^{(e)}\, \mathbf{F}_\Gamma\left(c_{r,s}^R(\beta_e), \beta_e \to \beta'_{r,s}\right) \mathbf{P}_{R,r,s}^{(e)}. \tag{110}$$

The shifted momentum $\beta'_{r,s}$ is Eq. (74), evaluated at $b = b_*$.

For an NRR theta graph the natural analytic variables are therefore

$$(c; h_N, \beta_1, \beta_2)\,, \tag{111}$$

with h_N, β_1, β_2 fixed in the c-recursion. A pole on one R edge shifts only its β; the other R momentum remains fixed even though its absolute conformal weight changes with the recursively evaluated central charge.

The remaining boundary datum is now the single limit

$$\mathbf{F}^{NRR}_{\Theta,\infty} = \lim_{\substack{c\to\infty \\ h_N,\beta_1,\beta_2 \text{ fixed}}} \mathbf{F}_\Theta(c; h_N, \beta_1, \beta_2). \tag{112}$$

This is simpler analytically than a multivariate large-h boundary problem, but it is not yet known: the R weights scale as $c/24$, so the seed is a hybrid large-c limit rather than the ordinary light superconformal block.

10.5 Sphere R-channel certificate

Before using the local kernel on a higher-genus graph, consider the mixed sphere block

$$\left\langle V_4^{\mathrm{NS}}(\infty)\, V_3^{\mathrm{R}}(1)\, V_2^{\mathrm{R}}(z)\, V_1^{\mathrm{NS}}(0)\right\rangle, \qquad \widehat{H}^{\eta_3,\eta_2}_{\beta}(c;q), \tag{113}$$

with an R intermediate module. The analytic variables are

$$(c; h_1, \beta_2, \beta_3, h_4; \beta, \eta_3, \eta_2)\,; \tag{114}$$

all variables after the semicolon are held fixed while c moves.

Canonical pole table. It is enough to retain the unordered labels $r > s \geq 1$, $r+s$ odd, and work on the positive sheet $\beta = \beta_{r,s}(b)$. Its two roots are

$$b^{(\alpha)}_{r,s} = \frac{\sqrt{2}\,\beta + \alpha\sqrt{2\beta^2 - rs}}{r}, \qquad c^{(\alpha)}_{r,s} = \frac{3}{2} + 3\left(b^{(\alpha)}_{r,s} + \frac{1}{b^{(\alpha)}_{r,s}}\right)^2, \qquad \alpha = \pm 1. \tag{115}$$

The description $\beta = -\beta_{r,s}(b)$ gives the same two c-poles after $b \mapsto -b$. Likewise, $r \leftrightarrow s$ is accompanied by $b \leftrightarrow b^{-1}$. Thus summing either of these equivalent descriptions would double count the poles. On the canonical sheet no endpoint-sign flip occurs; if the negative sheet is used instead, both η_2 and η_3 flip as in the published β-recursion.

At a stored root b_*,

$$\partial_c \beta_{r,s} = \left.\frac{r - sb^{-2}}{12\sqrt{2}\,(b + b^{-1})(1 - b^{-2})}\right|_{b=b_*}. \tag{116}$$

Consequently the coefficient residue is

$$\mathcal{R}^{(\alpha;\eta_3,\eta_2)}_{r,s} = -\frac{16^{rs/2} A^{(\beta),R}_{r,s}(b_*)}{\partial_c \beta_{r,s}(b_*)} P^{\eta_3}_{r,s}(\beta_3, h_4; b_*)\, P^{\eta_2}_{r,s}(\beta_2, h_1; b_*). \tag{117}$$

The shifted internal parameter is

$$\beta'_{r,s}(b_*) = (-1)^s \frac{rb_* - s/b_*}{2\sqrt{2}}. \tag{118}$$

The necessary regularization at $c = \infty$. The raw HJS elliptic series is not bounded coefficient by coefficient at fixed β. Already at level one its regular polynomial is

$$[q]H_{\beta}^{\eta_3,\eta_2}|_{\text{reg}} = \frac{c}{3} + 8\beta^2 - 8\beta_2^2 - 8\beta_3^2 - 8h_1 - 8h_4 - \frac{1}{2}. \tag{119}$$

The leading powers through the first computed orders are generated by the Ramond oscillator factor

$$\mathcal{E}_R(c,q) = \left[\prod_{n=1}^{\infty} \frac{1+q^n}{1-q^n}\right]^{c/6}. \tag{120}$$

We therefore define $H = \mathcal{E}_R\,\widehat{H}$. Since $\mathcal{E}_R$ is entire in c and independent of the internal momentum and component labels, the pole kernel is unchanged. Its first three regular coefficients are exactly

$$S_0 = 1, \qquad S_1(\beta) = 8\beta^2 - 8\beta_2^2 - 8\beta_3^2 - 8h_1 - 8h_4 - \frac{1}{2}. \tag{121}$$

If

$$X = \beta^2 - \beta_2^2 - \beta_3^2 - h_1 - h_4, \tag{122}$$

the next coefficient obtained directly from the fixed-β large-c limit is

$$S_2(\beta) = 32X^2 - 16\beta^2 + 12\beta_2^2 + 12\beta_3^2 - 4h_1 - 4h_4 - \frac{3}{8}. \tag{123}$$

Both S_1 and S_2 are independent of (η_3, η_2). At arbitrary order the normalized recursion is

$$[q^n]\widehat{H}_{\beta}(c) = S_n(\beta) + \sum_{\substack{r>s\geq 1,\ r+s \text{ odd} \\ rs/2 \leq n}} \sum_{\alpha=\pm 1} \frac{\mathcal{R}_{r,s}^{(\alpha)}}{c - c_{r,s}^{(\alpha)}} [q^{n-rs/2}]\widehat{H}_{\beta'_{r,s}}\left(c_{r,s}^{(\alpha)}\right). \tag{124}$$

The endpoint component labels in (124) are unchanged on the canonical positive sheet.

Level-one certificate. In the basis $(L_{-1}w^+, G_{-1}G_0w^+)$, the exact Gram matrix is

$$B_1 = \begin{pmatrix} 2h & -\frac{3}{2}\beta^2 \\ -\frac{3}{2}\beta^2 & -\beta^2(2h + c/4) \end{pmatrix}, \qquad h = \frac{c}{24} - \beta^2. \tag{125}$$

Its nontrivial Kac polynomial is

$$K_1(c,\beta) = c^2 - 30\beta^2 c + 144\beta^4 + 81\beta^2. \tag{126}$$

The two zeros of K_1 are precisely $c_{2,1}^{(+)}$ and $c_{2,1}^{(-)}$. Contracting B_1^{-1} with the two Ward vectors, or instead using (124) with the two residues (117) and the seed (121), gives the same answer for all four (η_3, η_2). The numerical residue test approaches the predicted value with relative error below 2×10^{-9}. At q^2, the same recursion includes the shifted-β' level-one tail and the two $(4,1)$ poles. Together with (123), it reproduces the full HJS coefficient for all four component choices.

This completes the pole part of the sphere R-channel c-recursion. The remaining derivation problem is sharply isolated: determine the closed hybrid large-c seed $S_n(\beta)$ for $n \geq 3$. Substituting an ordinary scalar $\mathfrak{osp}(1|2)$ global block here would be unjustified because the R weights scale as $c/24$.

10.6 The fixed-β contracted Ramond algebra

The sphere calculation suggests that the regular block should itself be regarded as a conformal block for the algebra which survives at $c = \infty$. This "smaller algebra" is not literally a subalgebra obtained by discarding most super-Virasoro generators. It is an Inönü–Wigner contraction of the full Ramond algebra, together with a renormalized limit of its three-point intertwiners.

Starting algebra and fixed-β grading. In the ordinary-c convention,

$$\begin{aligned}
[L_m, L_n] &= (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}, \\
[L_m, G_n] &= \left(\frac{m}{2} - n\right) G_{m+n}, \\
\{G_m, G_n\} &= 2L_{m+n} + \frac{c}{3}\left(m^2 - \frac{1}{4}\right)\delta_{m+n,0}, \qquad m, n \in \mathbb{Z}.
\end{aligned} \tag{127}$$

The correct finite zero-mode operator in the fixed-β limit is

$$D = L_0 - \frac{c}{24}. \tag{128}$$

On the Ramond ground fiber $\mathcal{R}_\beta^{(0)}$,

$$D|\beta, a\rangle = -\beta^2 |\beta, a\rangle, \qquad G_0^2 = D, \tag{129}$$

where a is the open two-dimensional Ramond ground index. Thus both D and G_0 remain unscaled and finite as $c \to \infty$.

Rescaled nonzero modes. For every positive integer n, define

$$\begin{aligned}
a_n &= \sqrt{\frac{12}{cn^3}}\, L_n, \qquad & a_n^\dagger &= \sqrt{\frac{12}{cn^3}}\, L_{-n}, \\
\psi_n &= \sqrt{\frac{3}{cn^2}}\, G_n, \qquad & \psi_n^\dagger &= \sqrt{\frac{3}{cn^2}}\, G_{-n}.
\end{aligned} \tag{130}$$

These are normalized versions of the original superconformal modes, not new local fields. Here $\dagger$ denotes the creation-mode partner in the chiral BPZ pairing; no Hilbert-space reality condition is being imposed. The $c \to \infty$ statements below are understood coefficientwise on fixed-level descendants. On a state at finite level ℓ, $D = -\beta^2 + \ell$ remains $O(c^0)$; no claim is made about a simultaneous limit in which the descendant level grows with c.

Acting on the ground fiber, the normalized one-oscillator descendants are exactly

$$\begin{aligned}
a_n^\dagger |\beta, a\rangle &= \frac{L_{-n}}{\sqrt{cn^3/12}} |\beta, a\rangle, \\
\psi_n^\dagger |\beta, a\rangle &= \frac{G_{-n}}{\sqrt{cn^2/3}} |\beta, a\rangle.
\end{aligned} \tag{131}$$

Indeed, their unnormalized diagonal norms are

$$\begin{aligned}
\langle \beta, a | L_n L_{-n} | \beta, b\rangle &= \left(\frac{cn^3}{12} - 2n\beta^2\right) \langle \beta, a | \beta, b\rangle, \\
\langle \beta, a | G_n G_{-n} | \beta, b\rangle &= \left(\frac{cn^2}{3} - 2\beta^2\right) \langle \beta, a | \beta, b\rangle.
\end{aligned} \tag{132}$$

The factors in Eq. (130) are therefore the unique positive-mode normalizations, up to finite phases, which produce unit leading Gram matrix. At equal mode number the corresponding exact diagonal brackets are

$$
\begin{aligned}
[a_n, a_n^\dagger] &= 1 + \frac{24D}{cn^2}, \\
\{\psi_n, \psi_n^\dagger\} &= 1 + \frac{6D}{cn^2}.
\end{aligned}
\tag{133}
$$

On any fixed-level family of states, brackets at unequal mode number are suppressed by $c^{-1/2}$. The $c \to \infty$ oscillator algebra is therefore

$$
\begin{aligned}
[a_n, a_m^\dagger] &= \delta_{nm}, & [a_n, a_m] &= [a_n^\dagger, a_m^\dagger] = 0, \\
\{\psi_n, \psi_m^\dagger\} &= \delta_{nm}, & \{\psi_n, \psi_m\} &= \{\psi_n^\dagger, \psi_m^\dagger\} = 0,
\end{aligned}
\tag{134}
$$

with all boson–fermion supercommutators zero.

The grading operator is not central. Its surviving action is

$$
\begin{aligned}
[D, a_n] &= -n a_n, & [D, a_n^\dagger] &= n a_n^\dagger, \\
[D, \psi_n] &= -n \psi_n, & [D, \psi_n^\dagger] &= n \psi_n^\dagger.
\end{aligned}
\tag{135}
$$

Most importantly, the Ramond zero mode does not decouple:

$$
\begin{aligned}
[G_0, a_n] &= -\sqrt{n}\, \psi_n, & [G_0, a_n^\dagger] &= \sqrt{n}\, \psi_n^\dagger, \\
\{G_0, \psi_n\} &= \sqrt{n}\, a_n, & \{G_0, \psi_n^\dagger\} &= \sqrt{n}\, a_n^\dagger, \\
G_0^2 &= D, & [D, G_0] &= 0.
\end{aligned}
\tag{136}
$$

Equations (134)–(136) are the complete contracted Ramond algebra, which we denote by

$$
\mathfrak{A}_{\mathrm{R},\infty} = (\mathfrak{h}_{\mathrm{bos}} \oplus \mathfrak{h}_{\mathrm{fer}}) \rtimes \langle D, G_0 \rangle. \tag{137}
$$

It is an infinite super-Heisenberg algebra acted on by a common Ramond supercharge. In particular, every nonzero mode survives after the appropriate rescaling; the limit is simpler because their nonlinear mode-changing interactions vanish, not because only finitely many modes remain.

Fock realization of the contracted Ramond module. Let Γ_β be the odd operator on $\mathcal{R}_\beta^{(0)}$ satisfying

$$
\Gamma_\beta^2 = -\beta^2. \tag{138}
$$

The tensor product with the oscillator Fock space is graded, so Γ_β anticommutes with the $\psi_n, \psi_n^\dagger$ and commutes with the $a_n, a_n^\dagger$. A representation of $\mathfrak{A}_{\mathrm{R},\infty}$ is

$$
\begin{aligned}
N &= \sum_{n=1}^{\infty} n \left(a_n^\dagger a_n + \psi_n^\dagger \psi_n \right), \\
D &= -\beta^2 + N, \\
G_0 &= \Gamma_\beta + \sum_{n=1}^{\infty} \sqrt{n} \left(a_n^\dagger \psi_n + \psi_n^\dagger a_n \right).
\end{aligned}
\tag{139}
$$

The graded cross terms in G_0^2 vanish, and each oscillator pair contributes $n(a_n^\dagger a_n + \psi_n^\dagger \psi_n)$; hence $G_0^2 = D$ holds on the entire contracted module.

A diagonal oscillator basis is

$$|\boldsymbol{k},\boldsymbol{\varepsilon};\beta,a\rangle=\prod_{n\geq 1}(a_n^\dagger)^{k_n}(\psi_n^\dagger)^{\varepsilon_n}|\beta,a\rangle,\qquad k_n\in\mathbb{Z}_{\geq 0},\quad \varepsilon_n\in\{0,1\},\tag{140}$$

with oscillator level

$$|\boldsymbol{k},\boldsymbol{\varepsilon}|=\sum_{n\geq 1}n(k_n+\varepsilon_n).\tag{141}$$

Apart from the fixed ground-fiber metric, its Gram matrix is diagonal with bosonic norm $\prod_n k_n!$. This is the first reason the contracted block can be evaluated without constructing super-Virasoro Gram matrices.

The conformal block of the contracted algebra. Let $\rho_{L,\infty}$ and $\rho_{R,\infty}$ denote the two renormalized N–R–R three-point intertwiners adjacent to the internal R edge. Keeping the Ramond ground indices open, define

$$\begin{aligned}\mathbf{S}_\beta(q)&=\sum_{\boldsymbol{k},\boldsymbol{\varepsilon}}q^{|\boldsymbol{k},\boldsymbol{\varepsilon}|}\rho_{L,\infty}\left(|\boldsymbol{k},\boldsymbol{\varepsilon};\beta\rangle\right)\left(B_\infty^{-1}\right)_{\boldsymbol{k},\boldsymbol{\varepsilon}}\rho_{R,\infty}\left(|\boldsymbol{k},\boldsymbol{\varepsilon};\beta\rangle\right)\\&=\left\langle\mathcal{V}_{L,\infty}\right|q^N\left|\mathcal{V}_{R,\infty}\right\rangle.\end{aligned}\tag{142}$$

The primary propagation $q^{-\beta^2}$ has already been extracted. Equation (142) is the precise meaning of the "conformal block for the smaller algebra." For the sphere channel,

$$\mathbf{S}_\beta(q)=\lim_{\substack{c\to\infty\\ h_1,h_4,\beta,\beta_2,\beta_3\ \text{fixed}}}\widehat{\mathbf{H}}_\beta(c;q)=\sum_{n=0}^\infty\mathbf{S}_n(\beta)q^n.\tag{143}$$

The scalar component computed in Eqs. (121) and (123) is independent of the endpoint component labels through q^2.

Renormalized intertwiners are additional data. The contracted commutation relations alone do not determine $\rho_{L,\infty}$ and $\rho_{R,\infty}$. They must be obtained by putting the exact N–R–R Ward identities in the variables (130), holding all β's and the adjacent NS weight fixed, and then taking $c\to\infty$. The leading coherent sources grow as $\sqrt{c}$; their Gaussian sewing produces the universal factor $\mathcal{E}_R(c,q)$ in (120). Thus the canonical statement is about the sewn functional,

$$\left\langle\mathcal{V}_{L,\infty}\right|q^N\left|\mathcal{V}_{R,\infty}\right\rangle:=\lim_{c\to\infty}\mathcal{E}_R(c,q)^{-1}H_\beta(c;q).\tag{144}$$

Splitting the counterterm into a separate square root assigned to each vertex is a convention, whereas the combined sewn limit above is unambiguous.

Since different positive mode numbers form independent free super-oscillators, the contracted Ward identities are expected to reduce the block to a product of small mode kernels,

$$\mathbf{S}_\beta(q)=\langle v_L|\overrightarrow{\prod_{n=1}^{\infty}}\mathbf{K}_n(q^n)\left|v_R\right\rangle.\tag{145}$$

The arrow records the ground-fiber and fermionic ordering. Once $\mathbf{K}_n$ is derived, truncating (145) at total q-degree N gives the first N seed coefficients exactly; it is not a truncation of the original super-Virasoro Gram problem.

Derivation target. The next representation-theoretic task is therefore finite and concrete:

1. contract the two ordered N–R–R Ward identities using Eqs. (130) and (136);
2. isolate the coherent $O(c)$ overlap producing $\mathcal{E}_R$;
3. determine the finite ground-fiber kernels $\mathbf{K}_n$;
4. check that their product reproduces S_1 and S_2;
5. use the resulting exact product as the regular input at every call of the fixed-β c-recursion.

This approach addresses the accuracy problem directly: the regular block is derived as an exact contracted-algebra block and is not estimated by ordinary descendant-level truncation.

10.7 Complete fixed-β c-recursion recipe

The following data are sufficient to turn the pole identities into an algorithm. We state them for the NRR theta block, whose natural reduced chiral block is

$$\mathbf{F}_\Theta = \mathbf{F}_\Theta\left(c; h_N, \beta_1, \beta_2; q_N, q_1, q_2\right). \tag{146}$$

The reduction must first extract the primary propagation factors

$$q_N^{h_N - c/24}, \qquad q_1^{-\beta_1^2}, \qquad q_2^{-\beta_2^2}, \tag{147}$$

and fix the normalization of the two ordered R–R–NS tensors. All R ground indices are kept open until the final sewing or GSO contraction.

1. Meromorphy and the subtraction degree. At each fixed plumbing multidegree $\boldsymbol{n}$, one must prove that the reduced coefficient is meromorphic in c, with only Kac poles, and determine its growth at $c = \infty$. If it is bounded, then

$$\mathbf{F}_{\Theta,\boldsymbol{n}}(c) = \mathbf{U}_{\Theta,\boldsymbol{n}} + \sum_{p \in \mathcal{P}_{\boldsymbol{n}}} \frac{\mathbf{R}_{p,\boldsymbol{n}}}{c - c_p}. \tag{148}$$

If it grows polynomially, $\mathbf{U}_{\Theta,\boldsymbol{n}}$ must be replaced by the complete subtraction polynomial. This analytic statement, not merely the list of Kac zeros, is what closes a c-recursion.

2. Pole tables. For the NS edge, hold h_N fixed and solve

$$h_N = h_{r,s}^{\mathrm{NS}}(c_{r,s}^{\mathrm{NS}}(h_N)), \qquad r + s \text{ even}. \tag{149}$$

Use

$$J_{r,s}^{\mathrm{NS}}(h_N) = \left[-\partial_c h_{r,s}^{\mathrm{NS}}(c)\right]^{-1}_{c = c_{r,s}^{\mathrm{NS}}(h_N)}. \tag{150}$$

For an R edge e, hold β_e fixed and solve

$$\beta_e = \epsilon \beta_{r,s}(b_*), \qquad c^*_{e,r,s,\epsilon} = c(b_*), \qquad r + s \text{ odd}. \tag{151}$$

Store the chosen b_* together with the c-pole: it is needed to evaluate $A_{r,s}$, the fusion polynomials, and $\beta'_{r,s}$. Equivalent $b \leftrightarrow b^{-1}$ branches are canonicalized before the pole is inserted.

3. Universal local residue kernels. For a pole on the NS edge,

$$\operatorname*{Res}_{c=c_{r,s}^{\mathrm{NS}}(h_N)} \mathbf{F}_\Theta = q_N^{rs/2} J_{r,s}^{\mathrm{NS}} A_{r,s}^{\mathrm{NS}} \Sigma_{r,s}^{(L)} \Sigma_{r,s}^{(R)} \mathbf{F}_\Theta \left(c_{r,s}^{\mathrm{NS}}; h_N + \frac{rs}{2}, \beta_1, \beta_2 \right). \tag{152}$$

For a pole on R edge e,

$$\begin{aligned} \operatorname*{Res}_{c=c^*_{e,r,s,\epsilon}} \mathbf{F}_\Theta = q_e^{rs/2} J_{r,s}^R(\beta_e;\epsilon) A_{r,s}^{(h),R} \mathbf{P}_{r,s}^{(L,e;\epsilon)} \\ \times \mathbf{F}_\Theta \left(c^*_{e,r,s,\epsilon}; \beta_e \longrightarrow \beta'_{r,s}(b_*) \right) \mathbf{P}_{r,s}^{(R,e;\epsilon)}. \end{aligned} \tag{153}$$

All labels not displayed in the shifted argument are spectators. One may equivalently use the published β-residue coefficient and replace $J^R A^{(h),R}$ by $-A^{(\beta),R}/\partial_c \beta_{r,s}$.

4. Component and sewing ledger. Every recursive state must include

$$\begin{gathered} (\text{edge sector, half-edge order, } \pm \text{ three-form, ground parity,} \\ \text{spin lift, R orientation, remaining multidegree}). \end{gathered} \tag{154}$$

The published fusion polynomials determine the scalar entries, but the ledger determines which entries and phases occur. Odd Ramond plumbing matrices and GSO projectors are applied only after the open-index block has been assembled.

5. Regular boundary data. The one genuinely new representation-theory input is

$$\mathbf{U}_\Theta(\boldsymbol{q}) = \lim_{\substack{c\to\infty \\ h_N,\beta_1,\beta_2 \text{ fixed}}} \mathbf{F}_\Theta(c; h_N, \beta_1, \beta_2; \boldsymbol{q}). \tag{155}$$

It should be derived by contracting the large-c super-Virasoro oscillator algebra while keeping $G_0^2 = -\beta_e^2$ finite. A direct Gram/Ward calculation through the first few multidegrees is needed only to identify and certify this seed, not as the final evaluation method.

6. Triangular recursion and base cases. At coefficient (n_N, n_1, n_2), include a pole on edge e only when $rs/2 \le n_e$, and call the shifted block at $n_e - rs/2$. Set negative multidegrees to zero and fix the level-zero matrix from the chosen R–R–NS three-point normalization. Memoization keys must contain c, all shifted labels, component data, and the remaining multidegree.

7. Collision-safe evaluation. Distinct Kac labels or quadratic branches can yield the same c_p, especially at $b = 1$. Equal pole keys must be merged as exact partial-fraction or Laurent objects before substituting the physical central charge. The recursion is performed at generic long $\beta_e \neq 0$; the short quotient at $\beta_e = 0$ is a separate final limit.

8. Certification. The implementation should pass, in increasing order of cost:

1. direct Gram/Ward coefficients at the first two nonzero multidegrees;
2. the published sphere and torus R-channel recursions after the appropriate graph degeneration;
3. $b \leftrightarrow b^{-1}$ and $\beta \leftrightarrow -\beta$ component relations;
4. equality of the two torus two-point plumbing channels;

5. factorization and spin-structure checks after nonchiral sewing.

Consequently, the currently known package is

$$\boxed{\text{pole positions} + \text{Jacobians} + \text{null norms} + \text{fusion matrices} + \text{label shifts.}} \tag{156}$$

The missing package is

$$\boxed{\text{fixed-}\beta \text{ large-}c \text{ seed} + \text{fully explicit higher-genus component ledger.}} \tag{157}$$

10.8 Role of Virasoro-primary decomposition

Restricting a super-Virasoro module to Virasoro modules gives an infinite tower of Virasoro-primary families. This may help derive the large-c or large-weight seed by resumming known Virasoro genus-two blocks. It is unlikely to be an efficient final algorithm because it introduces an infinite sum, nontrivial branching three-point tensors, and spurious Virasoro poles that cancel only after complete supermultiplets are recombined.

11 Established results and open tasks

Object	Status	Comment
Virasoro sphere four-point h-recursion	Established	Elliptic seed equals one.
Virasoro torus one-point and necklace recursion	Established	Universal character prefactor and local Kac residues.
Arbitrary Virasoro plumbing graph	Established via c-recursion	Large-c seed is global block times vacuum factor.
$\mathcal{N}=1$ NS sphere and torus h-recursion	Established	Requires even/odd component bookkeeping.
Ramond sphere and torus h-recursion	Established in fixed components	G_0 mixing is encoded in parity relations, fusion polynomials, and explicit β prefactors.
Genus-two NNN theta block	Not closed	Local poles are known; the multivariate regular seed is not.
Genus-two NRR theta block	Not closed	Requires open R ground fibres, two G_0 sewings, coupled component recursion, and a regular seed.

12 Recommended immediate derivation

The shortest useful calculation is not a high-level descendant expansion. It is the following finite problem:

1. Choose an explicit ordered basis for the two independent R–R–NS chiral tensors.

2. Write G_0, $(-1)^F$, and the two spin-lift sewing maps as 2×2 matrices.

3. Derive the Ward-identity action of a null embedding on this tensor basis.

4. Determine whether the two R-edge residue matrices can be simultaneously reduced to scalar sectors.

5. Compute the joint large-β_1, β_2 leading tensor and decide whether the explicit factor $\beta_1\beta_2$ leaves a finite seed.

If step 4 succeeds, the published scalar Ramond h-recursion generalizes almost directly. If it fails, the correct object is a small matrix-valued recursion. In either case, step 5 decides whether h-recursion closes or whether genus-two c-recursion is necessary.

A Basis-independent form of Ramond sewing

Let $\mathcal{G}_\beta$ be the 1|1 Ramond ground fibre, B_0 its BPZ metric, and S_ϵ the spin-lift map. The fixed-odd-modulus sewing operator is

$$\mathcal{K}_{\mathrm{R}}(q, \alpha; \epsilon) = q^{L_0 - c/24}\, B^{-1}\, S_\epsilon\, e^{\alpha G_0}. \tag{158}$$

After Berezin integration,

$$\int d\alpha\, \mathcal{K}_{\mathrm{R}}(q, \alpha; \epsilon) = q^{L_0 - c/24}\, B^{-1}\, S_\epsilon G_0. \tag{159}$$

The order of S_ϵ, G_0, and the BPZ map is part of the definition. A scalar component recursion is a projection of (158), not a replacement for it.

B Notation dictionary

Symbol	Meaning	Warning
h or Δ	Conformal weight	Both notations occur in the literature.
$q(z)$	Elliptic nome	Not the same as a local sphere coordinate z.
$A_{r,s}$	Inverse null-norm slope	Its value depends on the differentiation variable.
$P_{r,s}$	Null-vector fusion polynomial	Ordering and parity conventions matter.
$\pi = \mathrm{e}, \mathrm{o}$	Superconformal component parity	Not a spin structure by itself.
$(\pm)$ in R blocks	Chiral R–R–NS tensor structure	Not identical to the ground-state parity label.
β	HJS Ramond momentum	For physical super-Liouville momentum, typically $\beta = iP/\sqrt{2}$.
$\beta'_{r,s}$	Momentum of shifted null submodule	It replaces β after taking a Ramond residue.

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